LABORATORY REPORT · MICRO/NANOFABRICATION

Copper Photolithography on Glass:

Positive-Resist Etching vs. Negative-Resist Lift-off

Item	Details
Project type	Side project (self-directed fabrication study)
Institution	Shahid Beheshti University
Period	M.Sc. period, 2023–2026
Supervisor	Prof. Majid Mohseni
Materials	Sputtered Cu on glass; S1813 (positive) and AZ nLOF 2020 (negative) photoresists
Techniques	UV contact photolithography, RF sputtering, $FeCl_3$ wet etching, lift-off, optical microscopy

1. Abstract

Sputtered copper films on glass were patterned by UV contact photolithography using two complementary routes. Route A is subtractive: copper is sputtered first, a positive resist (S1813) protects the regions to be kept, and the exposed copper is removed by $FeCl_3$ wet etching. Route B is additive: a negative resist (AZ nLOF 2020) is patterned on bare glass, copper is sputtered over it, and the resist is dissolved in hot NaOH so that the metal on top of it is lifted off. The complete process chain – Cu sputter deposition, resist coating, mask alignment, exposure, pattern transfer and optical inspection – was carried out by the author. Both routes yielded continuous Cu patterns that reproduce their respective masks. Route A showed straight, well-defined line edges with two local defects; Route B was verified visually only. Exact bake times, exposure doses and film thicknesses were not recorded, and resists were processed according to the manufacturers' standard protocols.

2. Objective

The aim was to fabricate micro-scale copper patterns on glass with two opposite resist tones and to compare the two pattern-transfer strategies (etching and lift-off) in terms of process flow, equipment, and the quality of the resulting patterns.

3. Theoretical Background

The tone of the photoresist determines which pattern-transfer route is most suitable.

- **Positive resist (S1813):** exposed areas become soluble and are removed in the developer. The remaining resist masks the Cu beneath it, which makes the resist well suited to **etching**: Cu is deposited first and the unprotected metal is etched away.
- **Negative resist (AZ nLOF 2020):** exposed areas cross-link during the post-exposure bake and remain, while unexposed areas dissolve. The developed sidewalls have an undercut (negative) profile that interrupts the metal film at the resist edge. This is the standard choice for **lift-off**: Cu is deposited over the patterned resist, and the resist is then removed together with the metal on top of it.

Because the two resists have opposite tones, each route required its own photomask (a positive and a negative mask).

4. Process Overview

Table 1 summarises the two routes.

Parameter	Route A – Etching	Route B – Lift-off
Patterning type	Subtractive	Additive
Substrate	Glass	Glass
Cu deposition	Self-operated sputtering, before lithography	Self-operated sputtering, after lithography
Adhesion layer	None	None
Photoresist	S1813 (positive)	AZ nLOF 2020 (negative)
Spin coater	vCOAT (Backer), 1000 rpm, 40 s	vCOAT (Backer)
Exposure	Contact UV, mask aligner with alignment camera	Same tool, electrode mask
Pattern transfer	Wet etching in $FeCl_3$	Resist removal in hot NaOH (Heidolph MR 3001)
Inspection	Optical microscopy	Visual inspection

Table 1. Comparison of the two patterning routes.

5. Equipment and Materials

- **Sputtering system:** SATALAB (Micro/Nano Fabrication) six-target sputtering chamber with rotating substrate holder, RF power, automatic tuning (matching) controllers and mass-flow-controlled Ar supply.
- **Spin coater:** vCOAT (Backer).
- **Exposure:** contact UV mask aligner with alignment camera and monitor, operated under yellow-light conditions.
- **Hotplates:** IKA RCT basic (soft bake); Heidolph MR 3001 hotplate stirrer (lift-off).
- **Resists and chemicals:** S1813 (positive), AZ nLOF 2020 (negative), $FeCl_3$ etchant, NaOH solution, DI water.
- **Substrates:** clean glass; no Ti/Cr adhesion layer.
- **Inspection:** optical microscopy.

6. Experimental Procedure

6.1 Cu sputter deposition (both routes)

Cu films for both routes were deposited by the author on the SATALAB multi-target sputtering system, running the full sequence from pump-down to deposition (Figure 1).

- **Pump-down:** roughing followed by high vacuum, reaching the 10^{-4} mbar range (1.6×10^{-4} mbar on the control screen, Figure 1d).
- **Process readings:** Ar flow of 16 sccm; substrate rotation setting of 200 (Figure 1d).
- Deposition power, time and film thickness were not recorded.



(a) Six-target cathode array, Cu target in centre

(b) Chamber with substrate holder

(c) Control screen during pump-down



(d) Control screen at high vacuum, 16 sccm Ar

(e) Chamber viewport during operation

Figure 1. *Cu sputter deposition on the SATALAB system.*

6.2 Route A – S1813 (positive) + $FeCl_3$ wet etching

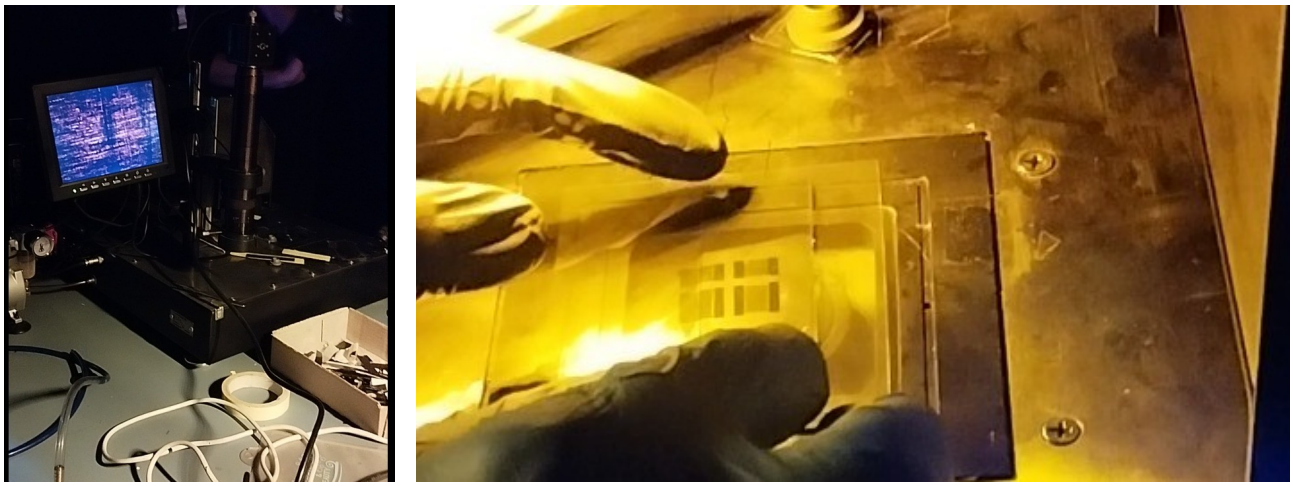
Cu was sputtered onto clean glass first; S1813 then defined the regions where Cu should remain, and $FeCl_3$ removed the remainder.

- **Cu deposition:** directly on clean glass substrates, with no Ti/Cr adhesion layer.
- **Resist coating:** S1813 spin-coated at 1000 rpm for 40 s on the vCOAT spin coater (Figure 2). From the manufacturer's spin curve this corresponds to roughly 2–2.5 µm of resist (estimate, not measured).
- **Soft bake:** on a hotplate following the S1813 standard protocol.
- **Alignment and exposure:** contact exposure under yellow light on a mask aligner with alignment camera and monitor (Figure 3).
- **Development:** exposed resist dissolved, leaving resist over the areas where Cu should remain.
- **Wet etching:** unprotected Cu etched in $FeCl_3$ solution (Figure 4).

- **Resist strip and inspection:** remaining resist removed and the pattern checked by optical microscopy (Section 7).



Figure 2. *S1813 spin coating on the vCOAT spin coater (display: 40 s, 1000 rpm).*



(a) Alignment camera and monitor (b) Mask and substrate under yellow light

Figure 3. *Route A – mask alignment and UV contact exposure.*

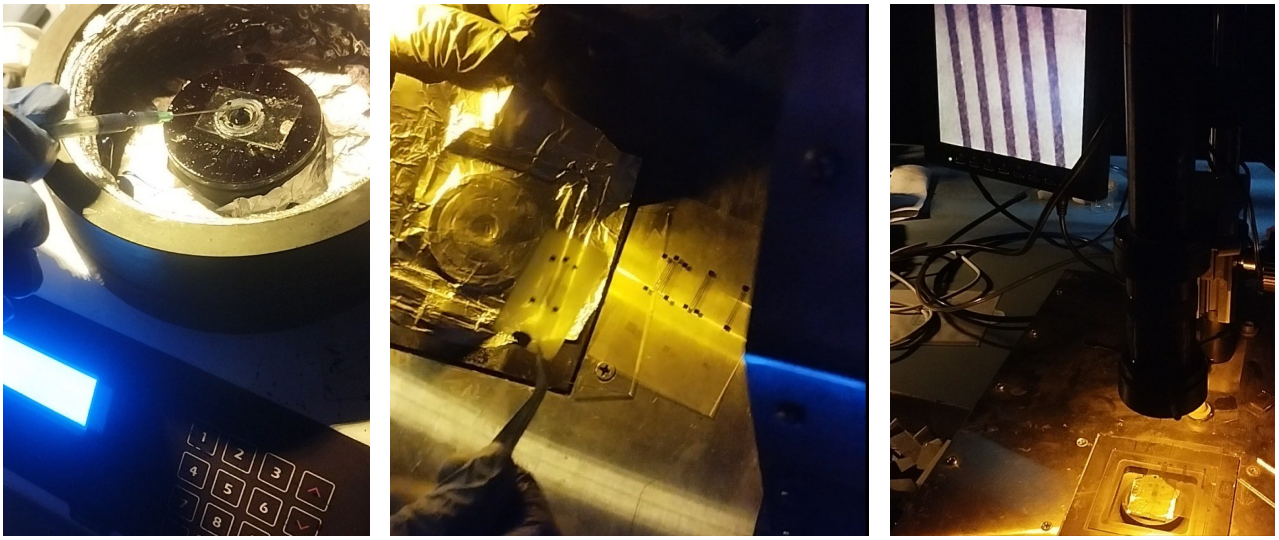


Figure 4. *Route A – Cu wet etching in $FeCl_3$ solution.*

6.3 Route B – AZ nLOF 2020 (negative) + lift-off

The resist was patterned on bare glass first; Cu was then sputtered over it, and the resist was removed together with the Cu on top of it.

- **Resist coating:** AZ nLOF 2020 dispensed by syringe onto bare glass and spin-coated on the vCOAT (Figure 5a).
- **Soft bake:** on an IKA RCT basic hotplate following the nLOF standard protocol.
- **Alignment and exposure:** the negative mask carries an electrode layout – groups of parallel lines with square contact pads, several devices per mask (Figure 5b–c).
- **Post-exposure bake and development:** cross-linking of the exposed resist, followed by development (Figure 6).
- **Cu deposition:** Cu sputtered over the patterned resist, with no Ti/Cr adhesion layer.
- **Lift-off:** because cross-linked nLOF dissolves poorly in acetone, the resist was removed in hot NaOH solution heated on a Heidolph MR 3001 hotplate stirrer (Figure 7), lifting off the Cu on top of it. $FeCl_3$ was used only for auxiliary cleanup of residual Cu.



(a) Resist dispensing by syringe (b) Mask under yellow light (c) Alignment monitor, line pattern

Figure 5. *Route B – resist coating and mask alignment/exposure.*



(a) Samples on foil-covered hotplate (b) Sample in beaker (wet processing)

Figure 6. *Route B – post-exposure bake and development.*



Figure 7. *Route B – hot NaOH lift-off on the Heidolph MR 3001 hotplate stirrer.*

7. Results and Discussion

Both routes produced continuous Cu patterns on glass that reproduce their masks.

Route A (etching). Optical microscopy (Figure 8) shows parallel Cu lines with straight, well-defined edges, indicating faithful pattern transfer from mask to film. Two local defects are visible: a particle on one line, and an irregular dark feature extending from a line edge into the gap, most likely residual unetched Cu or a resist defect. A feature of this kind can short adjacent electrodes.

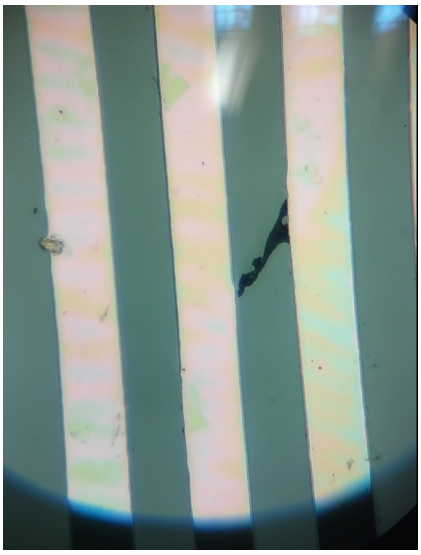


Figure 8. *Route A – optical micrograph of etched Cu lines showing a particle on one line and an irregular dark feature in the gap.*

Route B (lift-off). The electrode pattern of the negative mask was transferred to the glass after lift-off. No microscope image of this sample was recorded, so edge quality could not be compared directly with Route A.

8. Lessons Learned and Recommendations

Observation	Likely cause	Recommended improvement
No adhesion layer under Cu	Cu adheres poorly to glass and can peel during wet steps	Add 5–10 nm Cr or Ti before Cu
Particle and edge defect in Route A lines	Particle contamination; incomplete etching at one edge	Stricter substrate cleaning; controlled, timed etching with inspection
Hot NaOH used for lift-off	Low-cost substitute for commercial removers; may slightly oxidise Cu	Compare with NMP-based removers (e.g. TechniStrip)
No micrograph or thickness data for Route B	Not recorded at the time	Record microscopy and profilometry for every route to allow direct comparison

Table 2. Observations, likely causes and recommended improvements.

9. Conclusion

Etching with a positive resist and lift-off with a negative resist were both implemented successfully to pattern Cu on glass. The etching route gave well-defined line edges, with defects attributable to contamination and incomplete etching. The lift-off route transferred the electrode pattern but lacks microscopic verification. For future work, an adhesion layer, stricter cleaning, timed etching, and systematic recording of microscopy and thickness data for every route would allow a quantitative comparison of the two methods.

10. Skills Demonstrated

- UV photolithography with positive (S1813) and negative (AZ nLOF 2020) resists
- Spin coating, soft bake and post-exposure bake
- Mask alignment and contact exposure in a yellow-light environment
- Subtractive patterning by wet chemical etching ($FeCl_3$) and additive patterning by lift-off
- Sputter deposition of Cu thin films: vacuum pump-down, Ar process gas, target selection and substrate mounting (self-operated)
- Optical microscopy inspection and defect analysis